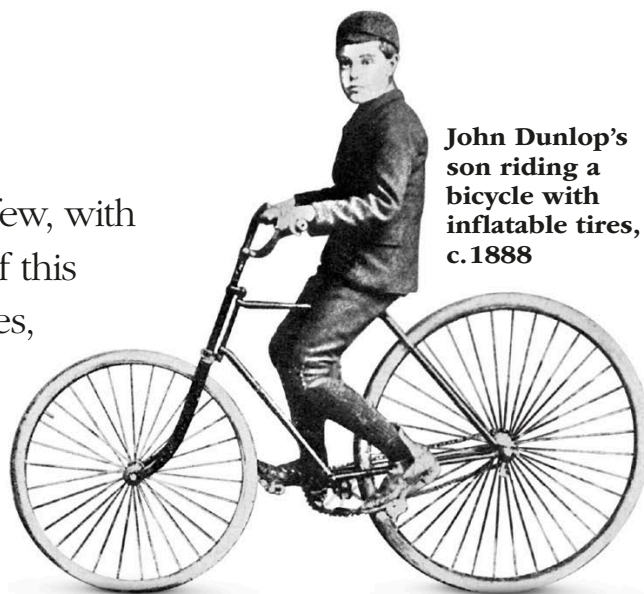


Bicycle

In the 1870s, riding a bicycle was for the brave few, with the rider perched on a giant wheel. The perils of this design led to the development of “safety” bicycles, which had a lightweight tube frame and two equal-sized wheels—a steerable front wheel and a chain-driven back wheel. The designs of bicycles today and the materials used to build them continue to be improved, while new technologies make cycling easier.



John Dunlop's son riding a bicycle with inflatable tires, c. 1888

INFLATABLE TIRE

Until 1888, people often rode bicycles with tires made of leather or solid rubber, resulting in a bumpy ride. Although the first inflatable rubber tire had been invented by the Scottish engineer Robert Thomson in 1845, it didn't really catch on until the Scottish inventor John Boyd Dunlop made the first practical, air-filled tire in 1887. He developed these tires after experimenting with rubber tubing—made from garden hose—on the wheels of his son's tricycle.



Chain attached to pedals drives the rear wheel.

Rear and front wheels are the same size.

ROVER SAFETY BICYCLE

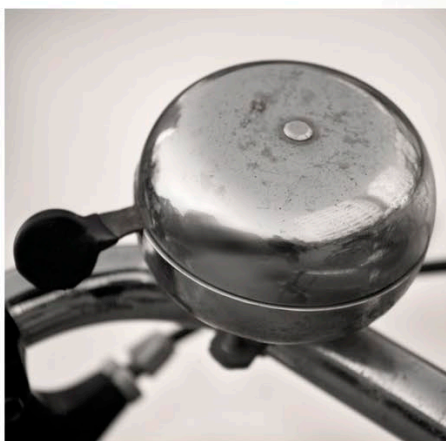
The safety bicycle was designed by the English inventor John Kemp Starley in 1885. With wheels no taller than a leg and a low, sprung saddle, the Rover Safety Bicycle was safer than other early bicycles, and more comfortable.

WOW!

In 2017,
more than
430 million people
owned bicycles
in China.

BICYCLE BELL

Around the same time Starley invented the Rover Safety Bicycle, the English inventor John Dedicoat is said to have invented the bicycle bell. It quickly became vital in those early days, when walkers had no experience of traffic except for noisy horses. Its basic design—a small, thumb-operated lever inside a round metal case—has remained popular.



Some helmets have streamlined tails to reduce air resistance.

